Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 101306 *Subject: Introduction to Civil Engineering* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer/ fill in the blanks of the following (Any seven only):* [2 x 7=14]

(a) Pneumatic Structure contains
(i) Metal (ii) Compressed air (iii) Wood (iv) Water

(b) A building can be mainly divided into ---------- components
(i) 1 (ii) 2 (iii) 4 (iv) 3

(c) Additional wall used to resist lateral forces (wind forces, earthquakes forces)
(i) Knee wall (ii) Infill wall (iii) Shear wall (iv) Cavity wall

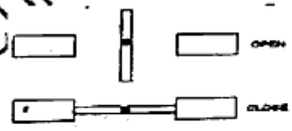
(d) Machine foundation is subjected to
(i) Static & dynamic Loads (ii) Kinematic Loads
(iii) Both (i) & (ii) (iv) None of the above

(e) Which IS code book guidelines for preparation of Mortar.
(i) IS 456-2000 (ii) IS 2250-1981
(iii) IS 3350-1981 (iv) None of the above

(f) While storing, a gap of 0.3 meter is to be maintained between cement bag and wall.

(g) The satellite Constellation of GPS consists of
(i) 4 satellites (ii) 6 satellites
(iii) 18 satellites (iv) 24 satellites

(h) The figure below shows mechanism of



(i) Sliding Door (ii) Spinning Door (iii) Swing Door (iv) Rotating door

(i) Usually, speed of Escalator is
(i) 25-30 m/min (ii) 10-25 m/min
(iii) 40-50 m/min (iv) 30-45 m/min

(j) What is the name of defect that will occur if paint is applied excessively thick?
(i) wrinkling (ii) Running
(iii) Flaking (iv) Grinning

Q.2 (a) List and explain, types of building classification as per national building code. [7]
(b) What is an ASPECT in building planning? Explain PROSPECT principles in building planning. [7]

Q.3 (a) With a neat sketch mention briefly about any two of modern surveying Instruments. [7]
(i) Electronic Distance Measurement
(ii) Total Station
(iii) Global Positioning

(b) Explain physical, mechanical properties & specifications of Indian standard bricks. [7]

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Q.4 (a) Difference between Pile and Pier Foundation. [7]

(b) Explain following Bearing capacity of soil. [7]

(i) Ultimate bearing capacity
(ii) Net safe bearing capacity
(iii) Gross safe bearing capacity
(iv) Net safe settlement pressure

Q.5 (a) Explain the principles of surveying. [7]

(b) What are different types of **Design Loads** in designing any structure. Also discuss about load combination. [7]

Q.6 (a) Mention name of two software from each domain and write their application in civil Engineering. [7]

(i) Drawing & drafting software
(ii) Design Software
(iii) Detailing Software
(iv) Project Management Software

(b) With a neat sketch and example (of Existing power plants in India), explain hydro power plant and thermal power plant working. [7]

Q.7 (a) Name typical ports in East and west coastal part of India and discuss their economical Important. Also list name of structures available at ports & harbour [7]

(b) Describe hydrological cycle with neat sketch and list various practical application of hydrology. [7]

Q.8 (a) Write short note on any two basic infrastructure services? [7]

(i) Air conditioning & purposes
(ii) Fire protection and materials
(iii) Ventilation, necessity and functional requirements.
(iv) Lifts & Escalators

(b) What are the open space requirements as per National Building Code norms? [7]

Q.9 Explain the following materials and list their various uses in building construction: [7x2=14

(a) Plain concrete
(b) Reinforced cement Concrete
(c) Prestressed concrete
(d) Ready mix concrete

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